Too Much Information? Mandarin Classifiers from Humans to LLMs

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SCAN ME!

1. Background

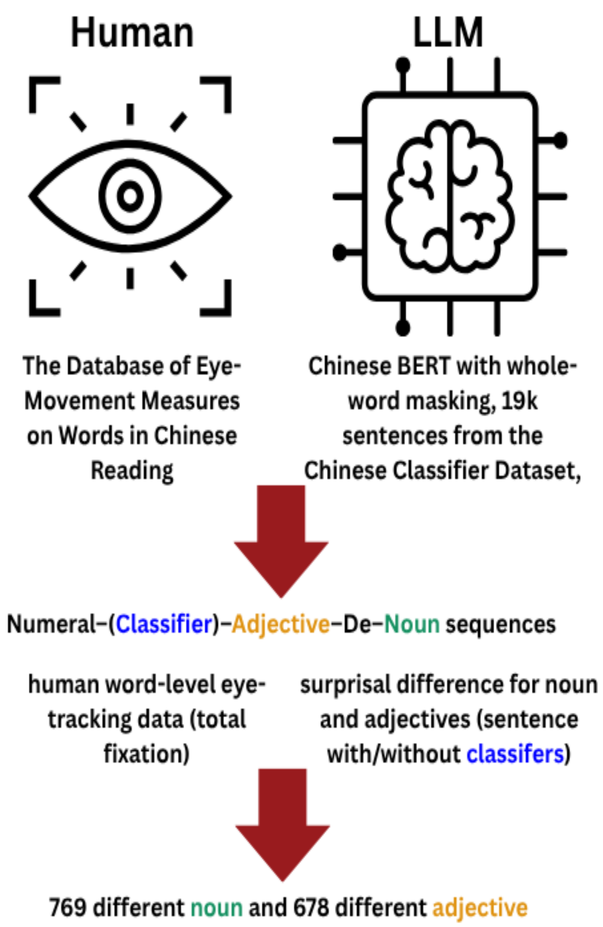
- Surprisal theory proposes that processing difficulty tracks a word's information content: words with higher surprisal are harder to process (e.g., Levy, 2008).
- Recent work has shown that surprisal estimates from large language models (LLMs) often correlate with human reading time, raising the question of whether LLMs process language in ways that resemble human comprehension (e.g., Merkx & Frank, 2021; Wilcox et al., 2020).
- Mandarin classifiers provide a useful test case for surprisal-based accounts and surprisal-distribution frameworks, based on their strict selection with nouns and more flexible selection with adjectives (e.g., Zhang, 2013)
- Previous studies in psycholinguistics have shown that classifiers reduce processing difficulty at the following noun, but their effects on following adjectives remain unclear (e.g., Hao et al., 2025).

2. Hypotheses

We investigate the following hypotheses:

1. Based on the Surprisal theory, classifiers should reduce surprisal for upcoming nouns and adjectives in the LLM data (and plausibly in human processing).
2. If LLM predictions mirror human comprehension, higher surprisal should be associated with shorter reading times in naturalistic reading

3. Dataset and Design



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4. Mandarin Classifiers

- Mandarin classifiers strongly constrain the noun they combine with: a classifier must be semantically compatible with the noun it classifies.

(1) yī tiáo gǒu
one CL.LONG dog
'one dog' (**Grammatical** match)

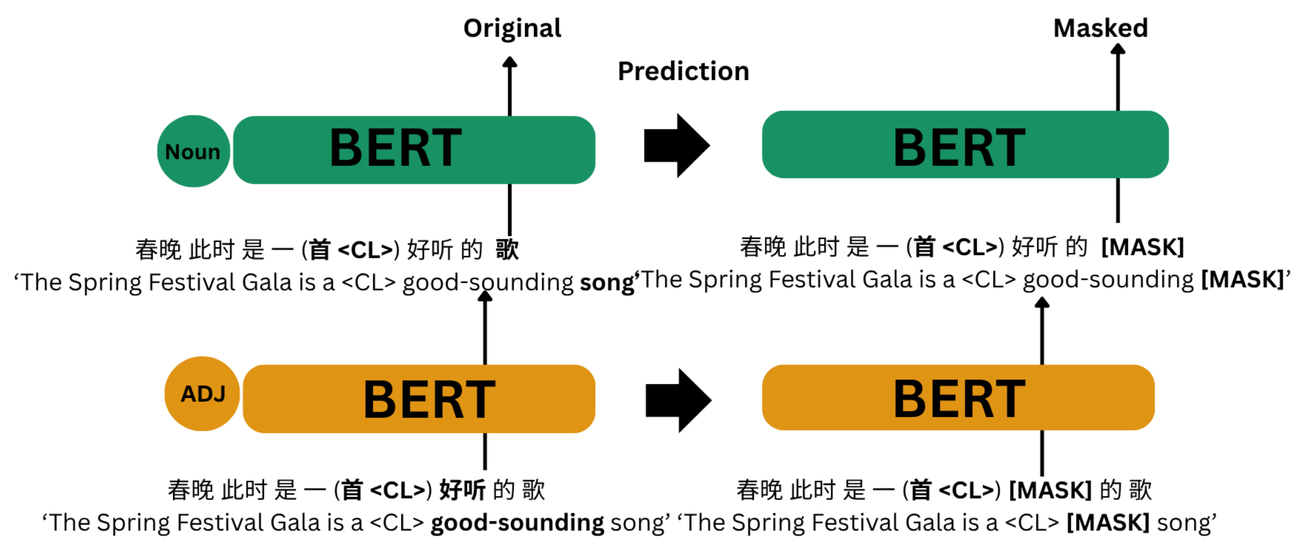
(2) *yī tiáo shū
one CL.LONG book
'one book' (**Ungrammatical**)

- They do **not** constrain following adjectives to the same extent: phrase acceptability depends mainly on classifier–noun compatibility, not on classifier–adjective compatibility.

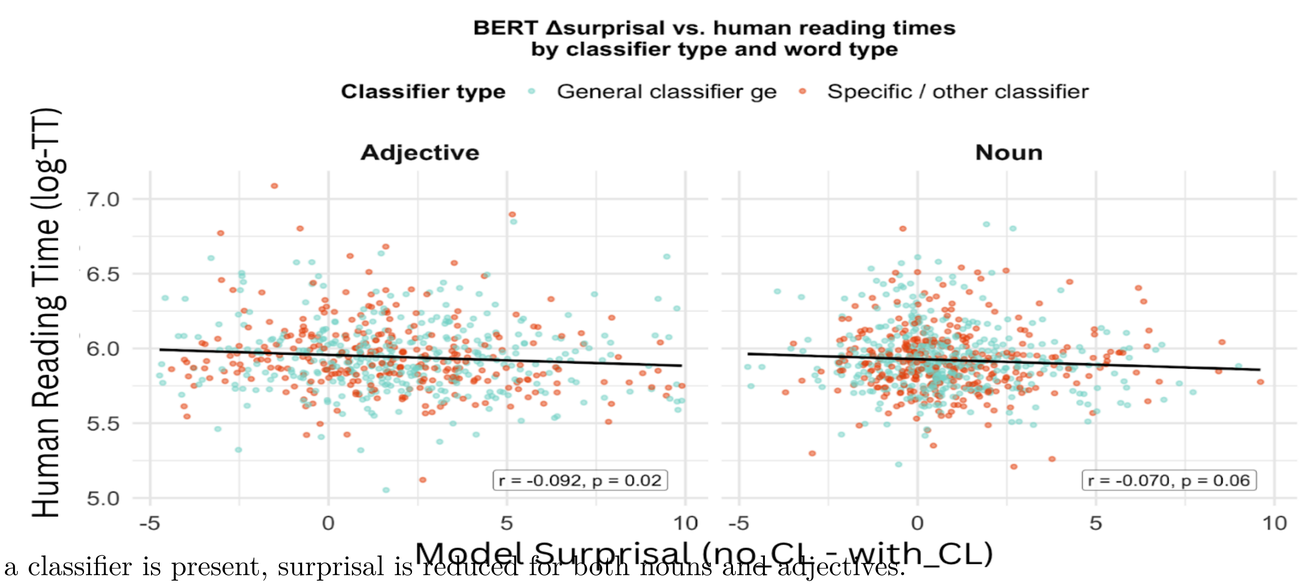
(3) yī tiáo dà gǒu
one CL.LONG big dog
'one big dog' (**Grammatical**)

(4) yī běn dà shū
one CL.FLAT big book
'one big book' (**Grammatical**)

5. Masked Language Modeling



6. Results



- When a classifier is present, surprisal is reduced for both nouns and adjectives.
- Modeling results were weakly negatively correlated with human data for both adjectives and nouns (marginal), significant for adjectives but only marginal for nouns

7. Take-home message

- **In LLMs**, Mandarin classifiers act as broad information-smoothing devices, reliably reducing surprisal for both upcoming nouns and adjectives.
- **In humans**, this pattern is only partially mirrored: words that are more helped by classifiers in LLMs are read slightly faster, but the effect is modest.
- LLM-based surprisal shows only **weak alignment** with human reading times, suggesting that surprisal alone does not fully explain human sentence processing.